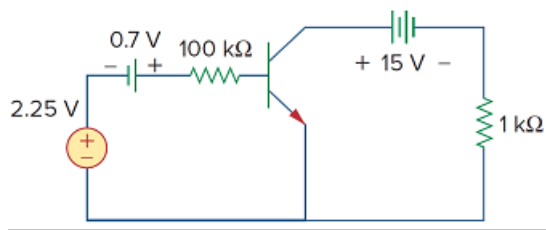


Problem

For the transistor circuit shown in Fig. 3.125, find I_B and V_{CE} . Let $\beta = 100$, and $V_{BE} = 0.7\text{ V}$.

Figure 3.125:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram.

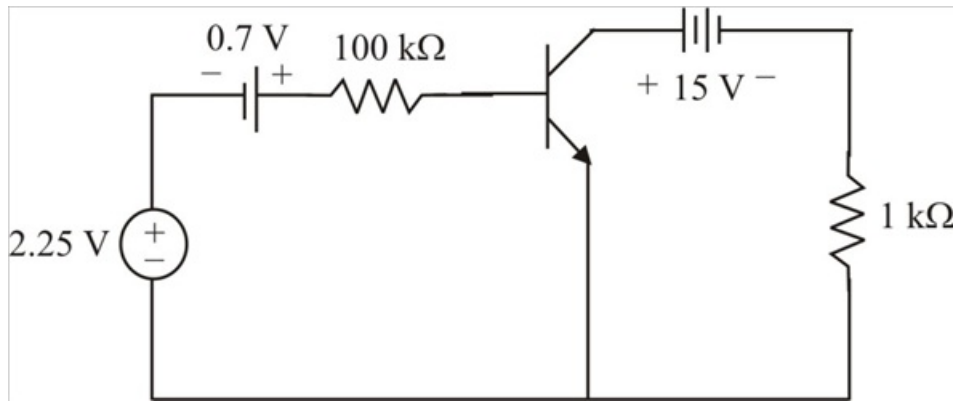


Figure 1

Step 2 of 4

The circuit is labeled and the Loops are shown in Figure 2.

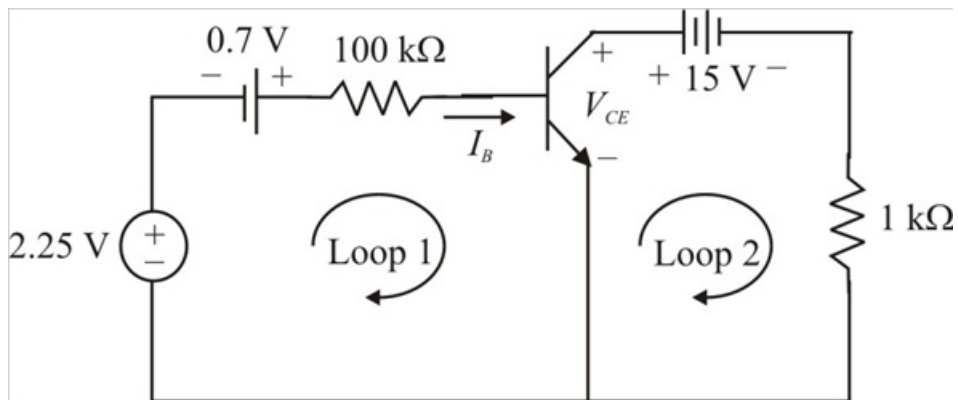


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage law in the loop 1.

$$-2.25 - 0.7 + (100 \times 10^3) I_B + v_{BE} = 0$$

$$-2.25 - 0.7 + (100 \times 10^3) I_B + 0.7 = 0$$

$$\begin{aligned} I_B &= \frac{2.25}{100 \times 10^3} \\ &= 22.5 \mu\text{A} \end{aligned}$$

Therefore, the current I_B is $22.5 \mu\text{A}$.

Step 4 of 4

Calculate the current I_C .

$$\begin{aligned} I_C &= \beta I_B \\ &= (100)(22.5 \times 10^{-6}) \\ &= 2.25 \text{ mA} \end{aligned}$$

Apply Kirchhoff's voltage law in the loop 2.

$$-V_{CE} + 15 + (1 \times 10^3)(-I_C) = 0$$

$$-V_{CE} + 15 - (1 \times 10^3)(2.25 \times 10^{-3}) = 0$$

$$\begin{aligned} V_{CE} &= 15 - 2.25 \\ &= 12.75 \text{ V} \end{aligned}$$

Therefore, the voltage V_{CE} is 12.75 V .